

Noninteger roots of a Schubert counting polynomial

Counterexample to TLMC 00000008430

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Abstract

The full Grassmannian of lines in a three-dimensional vector space over $\mathbb{F}_q$ has $q^2 + q + 1$ points. This is a Schubert configuration family determined by dimensions. Its counting polynomial is monic but has nonreal roots, disproving the assertion that its roots are integers.

1 Configuration family and scope

The source defines a Schubert configuration as a chain of subspaces determined by dimension conditions. It asserts that its finite-field count is a monic polynomial in q whose roots are integers.

Take $V = \mathbb{F}_q^3$ and the family of chains

$$0 \subsetneq L \subsetneq V, \quad \dim L = 1.$$

Each chain is uniquely determined by the line L , so this family is the Grassmannian $\mathrm{Gr}(1, 3) = \mathbb{P}^2$. It is the full Schubert variety, namely the closure of the largest Schubert cell for any fixed complete flag. The entire Grassmannian is allowed by the dimension-chain definition in the source. We count this full variety, not only its open affine cell, which by itself would have q^2 points.

2 Count and polynomial

Proposition 1. *The number of these configurations over every finite field of cardinality q is $q^2 + q + 1$.*

Proof. Every nonzero vector of V spans a unique line. Each line has exactly $q - 1$ nonzero vectors, and the nonzero-vector sets of distinct lines are disjoint. Thus the number of lines is

$$\frac{q^3 - 1}{q - 1} = q^2 + q + 1.$$

This holds for every finite field, not merely for selected small values. The polynomial $P(X) = X^2 + X + 1$ is also uniquely determined by these counts: any second counting polynomial agrees with it on every prime field cardinality, and there are infinitely many primes. Two distinct polynomials cannot agree at infinitely many points. $\square$

Proposition 2. *P is monic and has a root that is not an integer.*

Proof. Its leading coefficient is 1. Put

$$\zeta = \frac{-1 + i\sqrt{3}}{2}.$$

Direct expansion, using $(\sqrt{3})^2 = 3$, gives $\zeta^2 + \zeta + 1 = 0$. Since $\text{Im } \zeta = \sqrt{3}/2 > 0$, the root ζ is not a real number and in particular is not an integer. $\square$

The polynomial-counting and monicity clauses both hold in this example; it is precisely the integer-root clause that fails. Thus the example disproves the conjecture without imposing any alternate counting model.

3 Lean correspondence and verification

The Lean4.19.0 project pins Mathlib commit `c44e0c8ee63ca166450922a373c7409c5d26b00b`. For any field k , the type `Lines k` consists of actual `Submodule k` (`Fin 3 → k`) objects whose `Module.finrank` is one. Fixing the endpoints 0 and k^3 identifies these objects exactly with the chains above; no extra choices or quotients change the count.

The theorem `count_lines` applies Mathlib’s proved equivalence between projective points and one-dimensional submodules, followed by its projective-space cardinality theorem and the actual dimension of k^3 . It proves the count for every finite field. The theorem `polynomial_counts` identifies these cardinalities with evaluation of the actual complex polynomial `countPolynomial`. The theorem `counting_polynomial_unique` uses actual prime fields `ZMod p`, infinitude of primes, and the polynomial identity theorem to prove uniqueness of the counting polynomial.

The theorem `countPolynomial_monice` checks monicity. The value `zeta` is an actual complex number with the real square root of three in its imaginary part. The theorems `zeta_isRoot` and `zeta_not_integer` prove its root equation and exclusion from every complex cast of an integer. The final theorem `conjecture_00000008430` combines the general counting identity with existence of a noninteger root.

Run `lake build` and `lake env lean Main.lean` inside the `lean` directory. The axiom audits for the count, uniqueness, monicity, and final theorem use only the standard logical axioms `propext`, `Classical.choice`, and `Quot.sound`. No table of counts, unproved geometric assertion, native evaluation, numerical approximation, or auxiliary computational script is used.